

**College of Engineering**

Course	CC410 System Programming
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SHEET 8**SIC/XE Machine**

1. Trace The Following assembly Programs and translate it to object code (with and without reference numbers)

1 START prog1 0000 EXTDEF ALPHA, MAX EXTREF INF, ZERO LDS #3 LDT #300 LDX #0 CLOOP LDA ALPHA, X COMP MAX + JLT NOCH STA MAX NOCH ADDR S, X COMPR X, T JLT CLOOP ALPHA RESW 100 MAX WORD ZERO - INF	2 CSECT prog2 EXTDEF MIN, ZERO EXTREF ALPHA, MAX LDS THREE LDT #300 LDX #0 CLOOP +LDA ALPHA, X COMP MIN JGT NOCH STA MIN NOCH ADDR S, X COMPR X, T JLT CLOOP ZERO word 0 THREE word 3 MIN word MAX	3 CSECT prog3 EXTDEF INF EXTREF MAX, MIN, ALPHA +LDA MAX +LDS MIN DIVR S, A +STA ALPHA MULR S, A LDS BETA SUBR A, S STS DELTA ALPHA RESW 1 BETA RESW 1 GAMMA RESW 1 DELTA RESW 1 INF WORD 32768
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2. Consider the three above programs. Suppose they are linked in the sequence of PROG1+PROG2+PROG3, and the beginning load address is 001000. Derive the ESTAB that shows the external symbols and their addresses.
3. Show the memory content for prog1, prog2, prog3 after the operation of
- absolute loader
 - linking loader